

# Complementarity is Abundant, Predictability is Not the Problem:

## Decomposing Why LLM Routers Gain So Little

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### Abstract

LLM routers promise frontier quality at small-model cost. We ask where the promised gain actually goes. We decompose a router’s achievable gain over the correct baseline—a *cost-matched, query-independent* policy—into three multiplicative and additive terms: *complementarity*  $\kappa$  (how much oracle headroom the model set offers), *predictability*  $\rho$  (what fraction of that headroom is a function of the query at all), and *estimation error*  $\varepsilon$  (what a fitted router fails to capture). We prove  $\rho = 1$  exactly when outcomes are deterministic given the query, give a distribution-free ceiling  $\sqrt{\beta(1-\beta)} \text{sd}(\delta)$  on every router, and show  $\text{sd}(\delta)$  is identified only from *repeated* generations—which no public routing dataset provides. On RouterBench (36,494 prompts  $\times$  11 models, all 55 model pairs  $\times$  8 benchmark families) complementarity is abundant: median  $\kappa = 12.11$  pp of oracle headroom at matched cost. Yet the best of four router families captures 0.59 pp, or  $\rho_{\text{realised}} = 4.6\%$  of it. We then pre-registered and *refuted* our own explanation. Using  $k = 3$  replicate generations of 5,000 prompts across three models—data we generated because no public dataset replicates—we find per-item difficulty is highly *reliable* (0.77–0.98 of outcome variance is stable between-item signal, implying a Bayes-optimal AUC of 0.95–0.99), so  $\rho \approx 1$  and unpredictability is *not* the bottleneck. The entire gap is  $\varepsilon$ : which model will succeed is a stable property of the item that routers cannot read off the prompt. That gap survives nine router families—TF-IDF through a prompted 70B model and an *end-to-end fine-tuned* encoder—and a learning curve whose power-law asymptote is 0.74 AUC. Under a paired bootstrap with Holm correction, *no* family’s AUC advantage over a frozen MiniLM+logistic baseline is distinguishable from zero. Two findings sharpen the measurement problem itself: the highest-AUC router we built has the *worst* matched-cost gain (−1.31 pp), so per-model AUC is not a proxy for router value; and the whole analysis replicates on RouterBench’s independent 5-shot release ( $\kappa = 11.44$  pp,  $\rho_{\text{realised}} = 5.5\%$ ). Routing’s binding constraint is generalisation, not noise and not model redundancy.

## 1 Introduction

A cascade or router sends easy queries to a cheap model and hard ones to an expensive one, then reports savings against always using the expensive model. That comparison is nearly uninformative, because always-cheap and any fixed random mixture also beat always-expensive on cost while looking at nothing. The bar a router must clear is the best *query-independent* policy at the *same expected cost*. Reported gains against always-frontier are not evidence that reading the query helped.

Fixing the baseline is necessary but not sufficient, because it tells you *that* a gain is small without telling you *why*. Three very different worlds produce the same small number. The models might be redundant, so there is nothing to win. The outcome might be a coin flip given the query, so nothing is winnable by any router. Or the information might be there and the router might simply fail to extract it. The first is a model-selection problem, the second a property of the task, the third an engineering problem—and the correct response differs completely.

### Contributions.

1. **A decomposition that separates those three worlds (§2).** Achievable gain  $= \rho(b) \kappa(b)$ , and realised gain  $= \rho(b) \kappa(b) - \varepsilon$ . We prove the ladder  $S(b) \leq A^\dagger(b) \leq A^*(b)$ , closed forms for two models, a distribution-free ceiling on *every* router, the exact Bayes-optimal AUC implied by a difficulty

distribution, and—the key structural point—that  $\rho < 1$  is *exactly* outcome randomness given the query, so an oracle measured on single generations overstates attainable headroom. All propositions are validated numerically against brute-force LP and Monte-Carlo references (14/14 checks, Appendix B).

2. **Complementarity is abundant and routers capture almost none of it (§3).** On RouterBench, median  $\kappa = 12.11$  pp [11.15, 12.92] across 55 model pairs  $\times$  8 benchmark families; the best of four router families realises 0.59 pp [0.44, 0.78], i.e. 4.6% [3.6%, 5.9%] of it. The direction is consistent (49/55 pairs positive, 13/55 significant under Holm), so routing does work—it just recovers a twentieth of what is there.
3. **A pre-registered refutation of our own hypothesis (§4).** We predicted low predictability and tested it with  $k = 3$  replicate generations of 5,000 prompts  $\times$  3 models. The data says the opposite: difficulty is reliable,  $\rho \approx 1$ , and our ceiling bound is vacuous at these effect sizes. We report the refutation, and it sharpens the claim rather than weakening it.
4. **The gap is generalisation, and it does not close (§5).** Nine router families—TF-IDF, MiniLM+logistic, MiniLM+MLP, MiniLM+boosted trees,  $k$ -NN, random forest, a prompted 70B model zero-shot and 4-shot, and two *end-to-end fine-tuned* encoders—all land in 0.64–0.72 AUC, and none of their advantages over the frozen baseline survives a paired bootstrap with Holm correction. A learning curve over 250 to 29,000 training items extrapolates to an asymptote of 0.74.
5. **Per-model AUC is not a proxy for router value (§5).** The prompted 4-shot 70B router has the best AUC of any rung (0.710) and the *worst* realised matched-cost gain (−1.31 pp, i.e. worse than ignoring the query). AUC is invariant to the monotone rescaling of per-model scores that a matched-cost policy depends on, so a router can rank items well within each model and still make the wrong *cross-model* comparison on every item. The routing literature reports AUC or accuracy-at-a-budget almost universally; this is a concrete case where the two disagree in sign.

We claim no new router, architecture or algorithm, and we do not claim routing cannot work.  $\kappa$  and  $\rho$  are properties of a (workload, model set) pair; the contribution is the instrument that says which term is binding.

## 2 The decomposition

A query  $x \sim \mathcal{D}$ ;  $K$  models; running model  $m$  on  $x$  yields utility  $u_m(x) \in [0, 1]$  (graded correctness) and cost  $c_m(x) \geq 0$ . Both are random *given*  $x$ : decoding is stochastic, and measurably so even at temperature 0 (§4). Write  $\eta_m(x) = \mathbb{E}[u_m(x) \mid x]$  and  $\gamma_m(x) = \mathbb{E}[c_m(x) \mid x]$ . A policy assigns  $x$  to a model with probabilities  $z_m(x)$ , and has value  $A(z) = \mathbb{E}[\sum_m z_m(x) u_m(x)]$  and cost  $C(z) = \mathbb{E}[\sum_m z_m(x) c_m(x)]$ . Three nested classes, distinguished only by what they may look at, with optima at budget  $b$ :

$$S(b) \text{ (query-independent),} \quad A^\dagger(b) \text{ (router: any function of } x), \quad A^*(b) \text{ (oracle: sees realised } u, c).$$

**Proposition 1** (Ladder).  $S(b) \leq A^\dagger(b) \leq A^*(b)$ , and all three are concave and nondecreasing in  $b$ .

**Definition 1.**  $\kappa(b) = A^*(b) - S(b)$  (complementarity) and  $\rho(b) = (A^\dagger(b) - S(b))/\kappa(b)$  (predictability), so *achievable gain*  $= \rho(b)\kappa(b)$  and any fitted router  $\hat{z}$  realises  $A(\hat{z}) - S(b) = \rho(b)\kappa(b) - \varepsilon(\hat{z})$  with  $\varepsilon \geq 0$ .

The identity is true by construction; its use is that the three terms are separately measurable and assign blame differently ( $\kappa$ : the model set;  $\rho$ : the task;  $\varepsilon$ : the engineer).

**Proposition 2** (What  $\rho < 1$  is). *If  $u_m$  and  $c_m$  are deterministic given  $x$  for every  $m$ , then  $\rho(b) = 1$  for all  $b$ . Hence every unit of  $\rho < 1$  is outcome randomness given the query.*

Proposition 2 inverts how oracle headroom is normally read. A routing paper’s oracle is computed from *one* generation per model per  $x$  item, so it routes each item to whichever model happened to be right on that draw. When outcomes are noisy, part of what it collects is a coin flip it saw and no router can see. Published oracle headroom is an upper bound inflated by generation noise.

For  $K = 2$  (cheap  $s$ , dear  $l$ ,  $\beta$  = traffic fraction to  $l$ ), with  $\delta(x) = \eta_l(x) - \eta_s(x)$ ,  $D = u_l - u_s$ , and  $\text{CTE}_\beta$  the mean of the upper  $\beta$ -tail:  $S(b) = \mathbb{E}[u_s] + \beta\mathbb{E}[D]$ ,  $A^\dagger(b) = \mathbb{E}[u_s] + \beta\text{CTE}_\beta(\delta)$ ,  $A^*(b) = \mathbb{E}[u_s] + \beta\text{CTE}_\beta(D)$ , so  $\rho(\beta) = (\text{CTE}_\beta(\delta) - \mathbb{E}\delta)/(\text{CTE}_\beta(D) - \mathbb{E}D)$ : predictability is the ratio of tail dispersion of the *conditional mean* to that of the *realisation*. In the binary case the gain of an unconstrained oracle over the better single model is exactly  $\min(p_{01}, p_{10})$ .

**Proposition 3** (Router-free ceiling). *For every router at operating fraction  $\beta$ ,  $A^\dagger(b) - S(b) \leq \sqrt{\beta(1-\beta)} \text{sd}(\delta) \leq \frac{1}{2}\text{sd}(\delta)$ , and the bound is attained.*

*Proof.* For any  $z : \mathcal{X} \rightarrow [0, 1]$  with  $\mathbb{E}z = \beta$ ,  $\mathbb{E}[z\delta] - \beta\mathbb{E}[\delta] = \text{Cov}(z, \delta) \leq \text{sd}(z) \text{sd}(\delta)$  by Cauchy-Schwarz, and a  $[0, 1]$ -valued variable with mean  $\beta$  has variance at most  $\beta(1-\beta)$ . Tightness: take  $\delta$  two-valued and  $z$  its indicator.  $\square$

This is the same covariance object as the cost-accounting correction in §6: routing lives or dies on one covariance, read on two axes.

**Proposition 4** (Identification from replicates). *With  $k \geq 2$  conditionally i.i.d. replicates per item, binary outcomes, and independence across models,  $\text{Var}(\hat{\eta}_m) = \text{Var}(\eta_m) + \frac{1}{k}\mathbb{E}[\eta_m(1-\eta_m)]$  and  $\mathbb{E}[\frac{k}{k-1}\hat{\eta}_m(1-\hat{\eta}_m)] = \mathbb{E}[\eta_m(1-\eta_m)]$ , so  $\text{Var}(\delta)$  is identified by  $\widehat{\text{Var}}(\delta) = \text{Var}(\hat{\eta}_l - \hat{\eta}_s) - \frac{1}{k-1}[\hat{\eta}_l(1-\hat{\eta}_l) + \hat{\eta}_s(1-\hat{\eta}_s)]$ .*

$\kappa$  is computable from any released outcome matrix.  $\rho$  is not:  $\text{Var}(D) = \text{Var}(\delta) + \mathbb{E}[\text{Var}(D \mid x)] \geq \text{Var}(\delta)$ , so single-generation data bounds  $\text{sd}(\delta)$  only loosely, and *every public routing dataset stores one generation per (model, prompt)*. The quantity that decides whether routing can work is not recoverable from the data the field publishes. In simulation the uncorrected estimator inflates  $\text{Var}(\delta)$  by  $2.33\times$  (Appendix B).

**Proposition 5** (Bayes-optimal AUC). *With  $a = \mathbb{E}[\eta(X)]$  and  $X, X'$  i.i.d.,  $\text{AUC}^* = \frac{1}{2} + \mathbb{E}|\eta(X) - \eta(X')|/(4a(1-a))$ , and  $\text{AUC}^* \leq \frac{1}{2} + \text{sd}(\eta)/(2\sqrt{2}a(1-a))$ .*

Proposition 5 turns a replicate-estimated quantity into a prediction about a number we have measured with many learners—a check that cannot be tuned. Proofs: Appendix A.

### 3 Complementarity is abundant; routers capture 4.6% of it

**Setup.** RouterBench [Hu et al., 2024]: 36,494 prompts  $\times$  11 models = 401,434 inference outcomes with per-item graded score and per-item dollar cost, grouped into 8 benchmark families with  $n \geq 100$  (MMLU 14,042; HellaSwag 10,042; GSM8K 7,450; ARC 1,470; Winogrande 1,267; Chinese 785; other 931; MBPP 427). 21.1% of score cells are fractional partial credit, so we run the decomposition on graded utilities in  $[0, 1]$ , which needs no threshold. We verified these fractional scores are *not* replicate means—GPT-4 is modal at 0.75 and Mistral-7B at 0.25 with almost no mass at 0 or 1, and only one response is stored per model—so RouterBench cannot support Proposition 4.

We compute  $S(b)$  exactly (upper concave envelope of the per-model (cost, utility) points) and  $A^*(b)$  exactly (Lagrangian solution of the multiple-choice knapsack LP on *per-item* costs), for all 55 unordered model pairs and the full 11-model problem, over  $\beta \in \{0.1, \dots, 0.9\}$ ; headline numbers are at  $\beta = 0.5$ , where Proposition 3’s cap is largest and routing is therefore most favoured. Four router families are fit with 5-fold cross-validation grouped by item and stratified by family, so every router trains on 29,195 items. Routers commit using their predictions and per-model *mean* costs (a deployed router cannot know an item’s token count in advance) but are charged *realised* per-item cost.

Table 1 is the central empirical result. There is 12.11 pp of oracle headroom over a cost-matched query-independent policy, and the best router takes 0.59 pp of it. The effect is real: 49 of 55 pairs show a positive mean gain, 28 have raw  $p < 0.05$ , 26 survive Benjamini–Hochberg and 13 survive Holm–Bonferroni at family-wise  $\alpha = 0.05$ . So this is not a null result about routing; it is a statement about magnitude. Reading gains against always-frontier hides a factor of roughly twenty.

Note also that the MLP is the *worst* of the four despite the most capacity—the same inversion we observe in-house, and the first hint that the binding constraint is not model class.

Table 1: RouterBench,  $\beta = 0.5$ : medians over 55 model pairs  $\times$  8 benchmark families (440 cells), with 95% bootstrap intervals over cells.  $\rho_{\text{realised}}$  is realised gain divided by  $\kappa$ .

| Router                                     | out-of-fold AUC | gain (pp)                      | $\rho_{\text{realised}}$ | cells with gain $> 0$ |
|--------------------------------------------|-----------------|--------------------------------|--------------------------|-----------------------|
| TF-IDF + logistic                          | 0.7160          | +0.585 [0.442, 0.777]          | 0.046 [0.036, 0.059]     | 311/440               |
| MiniLM + logistic                          | 0.7081          | +0.455 [0.286, 0.609]          | 0.034                    | 288/440               |
| MiniLM + boosted trees                     | 0.6772          | +0.402 [0.268, 0.544]          | 0.033                    | 290/440               |
| MiniLM + MLP (256,64)                      | 0.6626          | +0.226 [0.138, 0.326]          | 0.017                    | 269/440               |
| <i>Complementarity <math>\kappa</math></i> |                 | <b>12.11</b> pp [11.15, 12.92] |                          |                       |

Table 2: Replicate-based decomposition,  $k = 3$ . *Reliability* is the share of per-item outcome variance that is stable between-item signal rather than regeneration noise. AUC\* is the Proposition 5 point estimate under a Beta fit to the replicate-identified moments.

| Set           | $n$   | Tier 1 (9B) |       | Tier 2 (20B) |       | Tier 3 (GPT-4o) |
|---------------|-------|-------------|-------|--------------|-------|-----------------|
|               |       | reliability | AUC*  | reliability  | AUC*  | reliability     |
| pool, $T=0.7$ | 5,000 | 0.895       | 0.992 | 0.873        | 0.987 | 0.942           |
| test, $T=0.0$ | 364   | 0.772       | 0.948 | 0.902        | 0.992 | 0.983           |

**Replication on the 5-shot release.** RouterBench ships a second, independent set of generations in which every prompt carries five in-context exemplars. It was pre-registered as a replication and analysed with identical code:  $\kappa = 11.44$  pp [10.66, 12.27], best-router gain +0.640 pp [0.519, 0.790],  $\rho_{\text{realised}} = 5.5\%$  [4.4%, 6.8%], with 50 of 55 pairs positive and 11 surviving Holm. Both hypotheses are supported again and the headline ratio moves by about a percentage point of  $\kappa$ . (28 of 36,508 5-shot items are dropped for missing score cells; the 0-shot file has none.)

## 4 Predictability is not the bottleneck: a refuted hypothesis

We pre-registered the hypothesis that  $\rho$  is small—that per-item success is weakly predictable in principle—and a falsification rule for it. To test it we generated  $k = 3$  replicates for every (item, model) over 5,000 prompts and three models at temperature 0.7 (45,000 calls) and 364 prompts at temperature 0 (3,276 calls), on HumanEval, MBPP, MMLU and GSM8K with execution- or exact-match grading. **The hypothesis was refuted.**

Table 2 says that whether a model answers an item correctly is not a coin flip: 77–98% of the outcome variance is a stable property of the item, and the implied Bayes-optimal AUC is 0.95–0.99. So  $\rho \approx 1$ : the information a router would need *is* there. Our pre-registered consistency check predicted the replicate-implied AUC\* would land within 0.05 of the 0.65–0.68 plateau measured by four learners; it came out 0.99 against 0.68, a miss of 0.31, and is recorded as inconsistent. We included that check because it could embarrass us, and it did.

Correspondingly Proposition 3’s ceiling is *vacuous* at these effect sizes: at  $\beta = 0.5$  it evaluates to 8.4–15.6 pp against a single-draw  $\kappa$  of 3.3–9.5 pp, a median ratio of 2.82. The inequality is correct and attained on a two-point distribution, but real  $\delta$  is diffuse, and Cauchy–Schwarz loses exactly that slack. We report this rather than reframing it.

Two things do survive. First, noise inflates the reported oracle: a policy allowed to peek at one graded outcome per model—information no deployed router has—captures only 34–55% of the single-draw  $\kappa$  when evaluated on held-out replicates. Second, 8–35% of the variance of the observed advantage  $D$  is generation noise, and the share scales with output length, from 8% for terse GPT-4o to 35% for a long-reasoning 9B model that changes its graded verdict on 19.8% of items between identical temperature-0 calls.

Table 3: Router-strength ladder on the same 1,500-item held-out split. Rows above the first rule are earlier stages of this project on identical data.  $\Delta\text{AUC}$  and its interval are a 2,000-resample paired bootstrap over items against the frozen MiniLM+logistic reference, Holm-corrected across the four new rungs. Gain is the realised matched-cost gain at  $\beta = 0.5$  against  $\kappa = 4.07$  pp on this split.

| Router                           | Representation         | mean AUC      | $\Delta\text{AUC}$ [95%] | gain (pp)      |
|----------------------------------|------------------------|---------------|--------------------------|----------------|
| $k$ -NN ( $k=50$ )               | MiniLM (frozen)        | 0.6521        | —                        | —              |
| Gradient boosting                | MiniLM (frozen)        | 0.6547        | —                        | —              |
| Random forest (train AUC 0.9999) | MiniLM (frozen)        | 0.6703        | —                        | —              |
| Logistic regression, Stage 7c    | MiniLM (frozen)        | 0.6816        | —                        | —              |
| Logistic regression, refit here  | MiniLM (frozen)        | 0.6896        | <i>reference</i>         | +0.556         |
| Prompted LLM, zero-shot          | Llama-3.3-70B-Instruct | 0.6416        | −0.048 [−0.097, +0.004]  | −0.444         |
| Prompted LLM, 4-shot             | Llama-3.3-70B-Instruct | <b>0.7105</b> | +0.021 [−0.019, +0.062]  | − <b>1.311</b> |
| Fine-tuned end-to-end            | MiniLM-L6 (unfrozen)   | 0.6945        | +0.005 [−0.014, +0.025]  | +0.533         |
| Fine-tuned end-to-end            | MiniLM-L12 (unfrozen)  | 0.6912        | +0.002 [−0.018, +0.023]  | +0.533         |
| LoRA fine-tune                   | Gemma-3-27B-it         | BLOCKED       | —                        | —              |

## 5 The binding term is generalisation

If  $\rho \approx 1$  and routers capture 4.6% of  $\kappa$ , then  $\varepsilon$  is essentially the whole story: difficulty is a stable property of the item that routers cannot read off the prompt. Two experiments ask whether that is fixable.

**More data?** A learning curve on RouterBench (fixed 20% held-out set, training subsets from 250 to 29,000 items, repeats at small sizes) gives held-out AUC  $0.65 \rightarrow 0.708$  for MiniLM+logistic. A power-law fit  $\text{AUC}(n) = A - Bn^{-c}$  gives asymptote  $A = 0.741$ , with  $\text{AUC}(10^6) = 0.726$  and  $\text{AUC}(10^9) = 0.738$ : roughly 2.5 doublings of data per +0.01 AUC. Realised gain does improve with data (+1.15 pp at 8k to +2.10 pp at 29k,  $\rho_{\text{realised}} = 0.105$ ), so data helps—but the extrapolated ceiling stays far below what capturing  $\kappa$  would need. (Exploratory; not pre-registered.)

**A stronger router?** Table 3 walks up nine router families on the identical 1,500-item held-out split. Nothing escapes the 0.64–0.72 band, and under a paired bootstrap over items with Holm–Bonferroni correction across the four new rungs, *none* of their AUC advantages over the frozen MiniLM+logistic reference is distinguishable from zero (smallest  $p_{\text{Holm}} = 0.26$ ). The largest point estimate, the prompted 70B model with four exemplars at +0.021 [−0.019, +0.062], is well below the +0.05 we pre-registered as the threshold at which “you tested a weak router” would become a live objection again.

Two rungs deserve comment. First, we *unfroze the encoder*: the same **all-MiniLM-L6-v2** whose frozen output gives 0.690 was fine-tuned end-to-end on the routing labels, which is the intervention that answers “maybe the representation is the problem.” It gains +0.005 [−0.014, +0.025]. Its inner-validation AUC reaches 0.751 while its held-out AUC is 0.694—so the capacity to fit routing labels exists and does not transfer, which is  $\varepsilon$  made visible.

Second, and this is the measurement point: **the highest-AUC rung has the worst matched-cost gain**. The prompted 4-shot router’s −1.31 pp is worse than ignoring the query entirely, at the same budget, on the same items. AUC is computed per model and is invariant to any monotone rescaling of that model’s scores, but a matched-cost policy ranks items by the *difference* between two models’ predicted utilities. A router can therefore order items correctly within each model and still get every cross-model comparison wrong. Since the routing literature reports per-model AUC or accuracy-at-a-budget almost universally, this is a concrete case where the standard metric and the quantity a deployment cares about disagree in sign.

The random forest is the other diagnostic: it reaches training AUC 0.9999—it memorises the training set perfectly—and still generalises *below* logistic regression. Ample capacity, no held-out gain. That is the signature of a target that is not a smooth function of the input representation, not of an inadequate model class.

**What we could not run, and what it costs the argument.** The pre-registered top rung was a fine-tuned *generative* LLM router. The LoRA job trained without incident (Gemma-3-27B-it, 3 epochs, 10,500 examples, 1,491,033 tokens, \$6.71 booked from the provider’s reported price) and then proved unservable through four independent routes, each probed rather than assumed: serverless LoRA inference is unavailable account-wide; dedicated endpoints v1 no longer accept creates platform-wide; the v2 resource model has no certified serving config for this base architecture; and re-training on a base that v2 *can* serve was refused for insufficient account balance. We report the rung as BLOCKED with those errors rather than estimating it, and we state the cost to the argument plainly: a reader who believes a fine-tuned generative LLM router would clear +0.05 has been answered by a fine-tuned encoder, a prompted 70B model, a memorising random forest and a learning curve—not by the thing they asked for.

## 6 Matched-cost baselines and a biased cost axis

Two measurement corrections underpin the numbers above and matter independently.

**The baseline.** On our own 364-item frozen test set, a deployed keyword router is beaten on *both* axes simultaneously by the constant policy “send everything to the mid tier”: 86.8% vs 92.3% accuracy at  $6.87\times$  the measured dollar cost. Its headline saving against always-frontier is 88.6% in modelled joules but 55.0% in measured dollars—the choice of cost axis moves the number by  $4\times$  on identical routing decisions. Applying the same audit to RouteLLM [Ong et al., 2025] on its own released GSM8K data, its router *does* beat a cost-matched query-independent mixture, by a mean of +0.57 pp: positive at 8 of 9 operating points (sign test  $p = 0.039$ ) but significant at none individually ( $n = 1,307$ ). So the protocol discriminates rather than rejects—and even the passing case has an effect the field’s benchmarks cannot resolve point-by-point, while being reported as “up to 85% cost reduction”.

**The axis.** Pricing a routing policy by multiplying traffic fractions by per-model *mean* cost is exact for a query-independent policy but biased for a router, because routers escalate precisely the items that emit more tokens. The exact correction is  $R_{\text{true}} = R_{\text{naive}} + \sum_{i \neq j} \text{Cov}(\mathbf{1}\{t^*=i, \hat{t}=j\}, e_j(x) - e_i(x))$ . On our data this term is  $1.9\times$  the regret being reported and flips its sign, reconciling to  $2 \times 10^{-16}$ ; on RouteLLM’s own GSM8K data the sign flip reproduces and naive accounting overstates savings by up to 6.3%. The bias always favours the router and grows with within-model cost dispersion, so it worsens as the field routes among reasoning models with variable-length thinking budgets. All routers in this paper are charged realised per-item cost.

## 7 Related work

RouteLLM [Ong et al., 2025] and FrugalGPT [Chen et al., 2023] report large savings against always-frontier; we re-audit the former against a cost-matched baseline. RouterBench [Hu et al., 2024] supplies the outcome matrix we decompose and proposes its own routing theory, but prices policies with per-model mean cost and does not separate complementarity from predictability. Concurrent evaluation-focused work reaches a compatible diagnosis from a different direction: Yuan et al. [2025] identify limited task diversity, imbalanced model pools and oversimplified metrics in existing routing benchmarks and build a larger framework (85 models, 33,337 queries), and Huang et al. [2025] likewise argue current router evaluation is inadequate. Our contribution is complementary and orthogonal: not a better benchmark, but a decomposition that attributes a measured shortfall to the model set, the task, or the router—plus the observation that the term which decides the question requires replicate generations that no such benchmark currently provides. The variance machinery is standard (reliability/ICC; CVaR [Rockafellar and Uryasev, 2000]; the mean–variance bound [Bhatia and Davis, 2000]); multiple-comparison control follows Holm [1979] and Benjamini and Hochberg [1995].



## 8 Limitations

$\rho$  is inferred, not bounded two-sidedly. Our rigorous ceiling is vacuous here; the claim  $\rho \approx 1$  rests on the reliability measurements and on a Beta fit to replicate-identified moments (Proposition 5 requires  $\mathbb{E}|\eta - \eta'|$ , which  $k = 3$  does not identify). The Beta fit is a modelling assumption and is labelled as such.

*Replicate assumptions.* Proposition 4 assumes replicates are conditionally i.i.d. and independent across models. Positive within-item correlation would understate the noise term and hence *overstate*  $\text{sd}(\delta)$  and the ceiling—erring in routing’s favour.

*Scale of the replicate study.* The replicate evidence is 48,276 calls over three models, four benchmarks and \$48.09 of real API spend. Three models is not a model ecosystem; the reliability numbers should be re-measured before being assumed elsewhere. The breadth comes from RouterBench, which cannot replicate; the depth from our own data, which is small. Neither alone suffices, and we do not claim the two settings are exchangeable.

*Benchmarks are not production traffic.* Single-turn, self-contained, auto-gradable academic tasks. Real routing traffic is conversational and mostly not objectively gradable. Every  $\kappa$  and  $\rho$  here is conditional on this item distribution.

*Energy is modelled, never measured.* Where earlier stages report joules they mean token counts times assumed per-tier rates; the headline comparisons in §6 are therefore stated in measured dollars.

*Generality.* Propositions 1, 2, 4 and 5 hold for general  $K$ ; Proposition 3 and the closed forms are two-model statements. The learning-curve extrapolation is a power-law fit, not a proof.

*The top rung of the ladder is missing.* The pre-registered fine-tuned *generative* LLM router is BLOCKED, for the four provider reasons in §5. Our strongest evidence against “you tested a weak router” is therefore an end-to-end fine-tuned encoder plus a prompted 70B model, not a fine-tuned generative router, and a reviewer is entitled to hold that specific objection open. Two things bound how much it can buy: the fine-tuned encoder has the *capacity* to fit the labels (inner-validation AUC 0.751) and the capacity does not transfer, and the effect would have to be  $10\times$  the largest we measured to change the conclusion.

*Null results with wide intervals are not proofs of absence.* Every  $\Delta\text{AUC}$  interval in Table 3 is roughly  $\pm 0.02$ – $0.05$  wide on 1,500 items. We can rule out the  $+0.05$  effect we pre-registered; we cannot rule out a  $+0.02$  one, and we do not.

## 9 Conclusion

Router gains decompose into complementarity, predictability and estimation error, and the three assign blame to the model set, the task and the router respectively. Measured across 11 models and 8 benchmarks, complementarity is abundant (12.11 pp) and routers capture 4.6% of it. We expected predictability to be the missing factor, pre-registered that hypothesis, and refuted it: per-item difficulty is highly reliable, so the information is there. What is missing is the ability to infer it from prompt text—a gap that nine router families, a 70B prompted model, two end-to-end fine-tuned encoders and a 29,000-item learning curve all fail to close. The practical implication is uncomfortable but actionable: for these workloads, effort spent on better routers is likely dominated by effort spent on choosing a more complementary model set, or on obtaining a signal that a prompt does not contain.

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## A Proofs

**Proposition 1 (ladder).** The inequalities follow from  $\Pi_0 \subset \Pi_1 \subset \Pi_2$ : a constant is a function of  $x$ , and a function of  $x$  is a function of  $(x, u, c)$ . For concavity,  $A$  and  $C$  are linear in  $z$  and each class is convex and closed under mixing, so if  $z^1$  is feasible at  $b_1$  and  $z^2$  at  $b_2$  then  $tz^1 + (1-t)z^2$  is feasible at  $tb_1 + (1-t)b_2$  and attains at least  $tA(z^1) + (1-t)A(z^2)$ . Monotonicity holds because enlarging  $b$  enlarges the feasible set.  $\square$

**Proposition 2 ( $\rho = 1$  iff outcomes are deterministic given  $x$ ).** If  $u, c$  are  $\sigma(x)$ -measurable then  $\Pi_1 = \Pi_2$  as policy classes with the same value and cost functionals, so  $A^\dagger = A^*$  and  $\rho \equiv 1$ .  $\square$

**Two-model closed forms.** Assume constant per-model costs  $c_s < c_l$  and  $\mathbb{E}[u_l] \geq \mathbb{E}[u_s]$ , and let  $\beta$  be the traffic fraction sent to  $l$ , so  $b(\beta) = (1-\beta)c_s + \beta c_l$ . (a) A query-independent policy mixes the two points and its value is linear and increasing in  $\beta$ , so the budget binds and  $S(b) = \mathbb{E}[u_s] + \beta\mathbb{E}[D]$ . (b) Maximising  $\mathbb{E}[u_s] + \mathbb{E}[z\delta]$  over  $z : \mathcal{X} \rightarrow [0, 1]$  with  $\mathbb{E}[z] \leq \beta$  is a fractional knapsack whose solution sets  $z = 1$  on the largest values of  $\delta$ , giving  $\beta\text{CTE}_\beta(\delta)$ . (c) Identical, with the oracle ranking by realised  $D$ . (d) Binary case: any policy has value at most  $\mathbb{E}[\max(u_s, u_l)] = 1 - p_{00}$ , attained by the cheapest-correct oracle; since  $\mathbb{E}[u_s] = p_{11} + p_{10}$  and  $\mathbb{E}[u_l] = p_{11} + p_{01}$  we get  $(1 - p_{00}) - \mathbb{E}[u_s] = p_{01}$  and  $(1 - p_{00}) - \mathbb{E}[u_l] = p_{10}$ , so the gain over the better single model is  $\min(p_{01}, p_{10})$ .  $\square$

Since  $\delta = \mathbb{E}[D \mid x]$  and  $\text{CTE}_\beta$  is a coherent risk measure, conditioning contracts it, which re-proves  $\rho \leq 1$  independently of Proposition 1.

Note that  $\min(p_{01}, p_{10})$  is *not*  $\kappa(b)$ : it measures headroom over the better single model, whereas  $\kappa(b)$  measures headroom over the matched-cost *mixture*, which is worse than the better single model whenever  $\beta < 1$ . A router can gain over a mixture even when  $\min(p_{01}, p_{10}) = 0$ , by spending a fixed budget where it does the most good. We report both because  $\kappa(b)$  is the quantity in the decomposition and  $\min(p_{01}, p_{10})$  is the quantity a reader intuit.

**Proposition 3 (ceiling).** In the main text. The variance step is the fact that a  $[0, 1]$ -valued random variable with mean  $\beta$  has variance at most  $\beta(1-\beta)$ , attained by the two-point  $\{0, 1\}$  distribution [Bhatia and Davis, 2000]. The maximum of  $\sqrt{\beta(1-\beta)}$  over  $\beta$  is at  $\beta = 1/2$ .

**Proposition 4 (identification).** The first identity is the law of total variance with  $\text{Var}(\hat{\eta} \mid x) = \eta(1-\eta)/k$ . For the second,  $k\hat{\eta} \sim \text{Bin}(k, \eta)$  given  $x$ , so  $\mathbb{E}[\hat{\eta}(1-\hat{\eta}) \mid x] = \frac{k-1}{k}\eta(1-\eta)$  and  $\frac{k}{k-1}\hat{\eta}(1-\hat{\eta})$  is unbiased for  $\eta(1-\eta)$ . Combining across the two models under conditional independence, with  $\frac{1}{k} \cdot \frac{k}{k-1} = \frac{1}{k-1}$ , gives the stated estimator.  $\square$



**Proposition 5 (Bayes AUC).** Score by  $\eta$ , which is AUC-optimal. With  $u = \eta(X)$ ,  $v = \eta(X')$  write  $T_{>} = \mathbb{E}[\mathbf{1}\{u > v\}u(1-v)]$ ,  $T_{<} = \mathbb{E}[\mathbf{1}\{u < v\}u(1-v)]$ ,  $T_{=} = \mathbb{E}[\mathbf{1}\{u = v\}u(1-v)]$ . Under the half-credit convention the AUC is  $(T_{>} + T_{=}/2)/(a(1-a))$ , and  $T_{>} + T_{<} + T_{=} = \mathbb{E}[u(1-v)] = a(1-a)$ . By exchangeability  $T_{<} = \mathbb{E}[\mathbf{1}\{u > v\}v(1-u)]$ , so  $T_{>} - T_{<} = \mathbb{E}[\mathbf{1}\{u > v\}(u-v)] = \frac{1}{2}\mathbb{E}|u-v|$ . Solving the two equations gives  $T_{>} + T_{=}/2 = \frac{1}{2}a(1-a) + \frac{1}{4}\mathbb{E}|u-v|$ , hence the closed form; the bound follows from  $\mathbb{E}|u-v| \leq \sqrt{\mathbb{E}(u-v)^2} = \sqrt{2}\text{sd}(\eta)$ . Sanity checks:  $\eta$  constant gives 1/2;  $\eta \in \{0,1\}$  equiprobable gives 1.  $\square$

## B Numerical validation of every proposition

Because the propositions are load-bearing, each is checked against a brute-force reference before any result depends on it (`stage11.13/s11.validate.py`, results in `s11.validate.json`). All 14 checks pass.

Table 4: Validation suite. “LP” references are `scipy.optimize.linprog` with HiGHS; others are Monte Carlo.

| Check                              | Reference                              | Result                                      |
|------------------------------------|----------------------------------------|---------------------------------------------|
| $S(b)$ exact                       | LP over the mixing simplex, 300 cases  | max err 0                                   |
| $A^*(b)$ exact                     | multiple-choice knapsack LP, 120 cases | max err 0, no budget overrun                |
| Ladder $S \leq A^\dagger \leq A^*$ | Bayes router vs. oracle, $n=20k$       | no violation                                |
| $\rho = 1$ iff deterministic       | analytic + Bernoulli noise             | exact; $\rho \rightarrow 0.258$ under noise |
| Ceiling never violated             | exact top- $\beta$ tail, 4,000 cases   | min slack $> 0$ ; max tightness 0.856       |
| Ceiling tight                      | two-point construction                 | ratio 0.998                                 |
| Variance components unbiased       | known Beta $\eta$ , 400 cases          | rel. bias 0.14%                             |
| $\text{sd}(\delta)$ unbiased       | known $\eta$ pair, 400 cases           | rel. bias 0.55%                             |
| Naive estimator inflated           | same simulation                        | $\text{Var}(\delta)$ inflated $2.33\times$  |
| Bayes AUC closed form              | empirical ROC AUC, 200 cases           | max err 0.0035                              |
| sd bound dominates                 | empirical AUC*                         | min slack 0.036                             |
| Beta moment match accurate         | empirical AUC*                         | max err 0.0030                              |
| $\min(p_{01}, p_{10})$ identity    | union minus best single                | max err 0                                   |
| Naive cost axis exact/biased       | simulated static vs. router            | 1.0% vs. 30.3% rel. error                   |

## C Pre-registration and deviations

Success criteria, falsification rules, the benchmark-family map, router families, statistics and the cost gate were committed to git *before* the analyses they govern (`stage11.13/prereg_stage11.13.md`); the commit precedes every result artifact and this can be checked against `git log`. This follows the same discipline as two earlier pre-registrations in the project.

Eight deviations are recorded in `stage11.13/DEVIATIONS.md`. The three that affect claims:

**D1.** H3 (“the router-free bound is  $\leq 50\%$  of  $\kappa$ ”) was **refuted**: the bound is  $2.82\times \kappa$  at median. The pre-registration named this outcome in advance and required us to retreat from the ceiling framing, which §4 does. The replacement claim — difficulty is reliable but not inferable from text — is the one the data supports.

**D2.** The pre-registration asserted that a policy seeing one replicate  $Y^{(1)}$  upper-bounds  $A^\dagger$ . *This is wrong*:  $Y^{(1)}$  is a realised outcome, not a function of  $x$ , so the two information sets are incomparable; only their join contains  $\sigma(x)$ . The error was caught while implementing Stage 12, before any result depended on it. The quantity is still reported, relabelled as the unbiased value of a feasible peeking policy rather than a bound.

**D8.** The pre-registered fine-tuned generative LLM rung is BLOCKED; Appendix F gives the four provider errors. Two additions are exploratory and labelled as such throughout: the learning curves (D3) and the end-to-end fine-tuned encoders (D7). The encoders are judged against the *same* pre-registered +0.05 threshold as the pre-registered rungs, so adding them cannot make the criterion easier to pass.

## D Per-family RouterBench detail

Full per-pair, per-family, per- $\beta$  output is in `stage11_13/s11_routerbench_pairs_0shot.csv.gz` (440 pair-family cells  $\times$  9 operating points, plus the 11-model problem). Summary statistics and all hypothesis verdicts are in `s11_routerbench_0shot.json`. The 5-shot file is analysed identically as a pre-registered replication and reported in `s11_routerbench_5shot.json`.

Multiple-comparison detail at  $\beta = 0.5$  for the best router: 49/55 pairs have a positive mean gain, 28 have raw  $p < 0.05$ , 26 survive Benjamini–Hochberg at FDR 0.05, and 13 survive Holm–Bonferroni at family-wise  $\alpha = 0.05$ .  $p$ -values are two-sided bootstrap over benchmark families with the router and baseline evaluated on identical items.

Table 5: The 5-shot release as a pre-registered replication of the 0-shot primary analysis, run with identical code. Medians over the same 440 pair-family cells at  $\beta = 0.5$ , 95% bootstrap intervals over cells.

| Quantity                      | 0-shot (primary)         | 5-shot (replication)               |
|-------------------------------|--------------------------|------------------------------------|
| items analysed                | 36,494                   | 36,480 (28 dropped, missing cells) |
| complementarity $\kappa$      | 12.11 pp [11.15, 12.92]  | 11.44 pp [10.66, 12.27]            |
| best router                   | TF-IDF + logistic        | TF-IDF + logistic                  |
| out-of-fold AUC               | 0.7160                   | 0.7172                             |
| realised gain                 | +0.585 pp [0.442, 0.767] | +0.640 pp [0.519, 0.790]           |
| $\rho_{\text{realised}}$      | 4.6% [3.6, 5.9]          | 5.5% [4.4, 6.8]                    |
| pairs with positive mean gain | 49/55                    | 50/55                              |
| pairs surviving BH / Holm     | 26 / 13                  | 22 / 11                            |
| H1, H2 verdicts               | supported                | supported                          |

## E Cost accounting for the new experiments

The project’s earlier stages spent \$48.09 of real API budget. Stage 13 was pre-registered with a \$25 gate on additional spend, enforced in code: every call accumulates the provider’s own reported usage and the run aborts if the gate is crossed (`stage11_13/s13_spend.json`).

Table 6: Additional spend for the new experiments. Every line is the provider’s own reported usage or job price, accumulated at call time.

| Item                                        | Volume                            | USD         |
|---------------------------------------------|-----------------------------------|-------------|
| Prompted router, zero-shot (Llama-3.3-70B)  | 4,500 calls                       | 0.58        |
| Prompted router, 4-shot (Llama-3.3-70B)     | 4,500 calls                       | 2.41        |
| LoRA fine-tune (Gemma-3-27B-it, 3 epochs)   | 10,500 examples, 1,491,033 tokens | 6.71        |
| Fine-tuned generative router inference      | BLOCKED, App. F                   | 0.00        |
| End-to-end fine-tuned encoders (CPU)        | 2 backbones $\times$ 6 epochs     | 0.00        |
| RouterBench (both releases), curves, theory | —                                 | 0.00        |
| <b>Total additional</b>                     |                                   | <b>9.71</b> |

The pre-registered gate was never the binding constraint: spend stopped at \$9.71 of \$25 because the provider refused further fine-tuning jobs, not because we chose to stop. RouterBench, the decomposition, the learning curves, the end-to-end encoder runs and all numerical validation cost nothing: they analyse already-released outcomes and already-paid-for generations, or are CPU work. The expensive part of this project is the replicate data, and it is the part no public dataset provides.

## F The blocked rung, in full

The pre-registration commits us to reporting a blocked run as blocked, with the provider’s error, and to estimating nothing in its place. Each route below was probed rather than assumed; verbatim responses are

stored in `stage11_13/s13_ftblocked.json` and `stage11_13/s13_endpoint_probe.json`.

1. **The fine-tune itself succeeded.** Job `ft-6567602b-4aa3`, LoRA on `google/gemma-3-27b-it`, 3 epochs over the 10,500 prompt/completion pairs in `s13_ft_train.jsonl` (the exact file is committed), 1,491,033 tokens, \$6.71.
2. **Serverless LoRA inference: unavailable.** `/v1/models` advertises 13 `*-Lora` inference targets including `google/gemma-3-27b-it-lora`; every one returns 400 `Unable to access non-serverless model` on this account, and the fine-tuned name itself returns 404 `model_not_available`.
3. **Dedicated endpoints v1: retired.** `POST /v1/endpoints` returns 403 `endpoints_v1_create_access_disabled` — a platform-wide change, not an account limit.
4. **Dedicated endpoints v2: no config for this base.** v2 requires a certified serving config; `models.configs.list` returns `zero` for both the merged fine-tune and the `gemma-3-27b-it` base, and that base is absent from the 43 v2-supported architectures. A `validate_only` deployment create fails for want of a config.
5. **Re-training on a servable base: refused.** Qwen/Qwen3.5-9B is fine-tunable *and* has a certified v2 config (1×H100 BF16, \$3.99/replica-hour), and it is the Tier-1 model of the system under study, so the router would have cost no more to run than the cheapest tier it routes to. `POST /v1/fine-tunes` returns 402 `insufficient_balance`: “Required combined balance and credit limit: 4.00 USD”. Pay-as-you-go serverless calls still return 200, so the block is the upfront reserve a fine-tuning job requires, not the credential.

## G Reproduction

```

pip install -r requirements-eval.txt
python3 stage11_13/s11_validate.py           # 14/14 proposition checks
curl -L -o external_data/routerbench_0shot.pkl \
  https://huggingface.co/datasets/withmartian/routerbench/resolve/main/routerbench_0shot.pkl
python3 stage11_13/s11_routerbench.py 0shot  # decomposition, 55 pairs x 8 families
python3 stage11_13/s12_ceiling.py         # replicate variance components
python3 stage11_13/s11_routerbench.py 5shot # pre-registered replication
python3 stage11_13/s12b_learning_curve.py  # learning curves
python3 stage11_13/s13b_encoder_finetune.py # end-to-end encoder rungs, CPU, $0
python3 stage11_13/s13c_encoder_routerbench.py 0shot # same, on RouterBench
python3 stage11_13/s13_llm_router.py probe  # cost projection before spending
python3 stage11_13/s13_llm_router.py report  # C1/C2 verdicts and bootstrap CIs

```

Seeds, package versions, model strings, endpoints and dataset revisions for all stages are listed in `stage7_10/reproducibility_manifest.md` and `stage11_13/`. Reproducibility hazards are flagged rather than glossed: no provider exposes an immutable model revision, so exact regeneration of any API-derived number is guaranteed by nothing under our control. The RouterBench pickle is pinned by SHA-256 in the result JSON.